

FILTER DESIGN TEST CASE

FBAR (Film Bulk Acoustic Resonator) - Band 1 (2100 MHz) for UWB (Ultra-Wideband) Filter
Design Validation and Performance Characterization

DOCUMENT INFORMATION

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Priority:	P1 - Critical

1. OBJECTIVE

This document describes the design, simulation, and characterization of a FBAR (Film Bulk Acoustic Resonator) filter targeting Band 1 (2100 MHz) for UWB (Ultra-Wideband) Filter. The filter is designed on AIN on Si substrate with a target center frequency of 5819.2 MHz and bandwidth of 297.3 MHz. The objective is to validate that the filter meets the required insertion loss, rejection, and linearity specifications for integration into the UWB (Ultra-Wideband) Filter module. This design iteration addresses feedback from the previous revision regarding out-of-band rejection improvements and temperature stability.

2. SCOPE

- Filter Type: FBAR (Film Bulk Acoustic Resonator)
- Frequency Band: Band 1 (2100 MHz)
- Application: UWB (Ultra-Wideband) Filter
- Substrate: AIN on Si
- Package: Flip-Chip BGA
- Design Tool: CST Studio Suite 2024
- Design Center: Andover HQ Lab

This specification applies to all variants of the FBAR (Film Bulk Acoustic Resonator) design targeting Band 1 (2100 MHz). Results are used to qualify the design for mass production release.

3. DESIGN PARAMETERS

Center Frequency:	5819.2 MHz
Bandwidth (3 dB):	297.3 MHz
Insertion Loss Target:	≤ 3.7 dB
Return Loss Target:	≥ 17.0 dB
Out-of-Band Rejection:	≥ 24.0 dB
Isolation (Duplexer):	≥ 44.0 dB
Q Factor:	3130
Temperature Coefficient:	-25.6 ppm/°C
Die Size:	0.6 x 1.5 mm
Wafer Size:	8-inch
Number of Resonators:	4
Electrode Material:	Al

Passivation: SiO2/Si3N4 stack

4. TEST PROCEDURE

1. Open CST Studio Suite 2024 and load the FBAR (Film Bulk Acoustic Resonator) template project.
2. Define the substrate stack: AlN on Si with specified layer thicknesses.
3. Set design targets: center freq 5819.2 MHz, BW 297.3 MHz.
4. Run initial EM simulation with 2D mesh density of 20 cells/wavelength.
5. Optimize resonator geometry using gradient descent (target IL < 3.7 dB).
6. Verify coupling coefficients between resonator stages.
7. Run 3D FEM simulation to account for package parasitics (Flip-Chip BGA).
8. Export GDS-II layout for mask fabrication.
9. Fabricate prototype wafers (8-inch, AlN on Si).
10. Perform on-wafer probing using Rohde & Schwarz ZNB40.
11. Measure S-parameters (S11, S21, S12, S22) from 100 MHz to 17458 MHz.
12. Compare measured results against simulation and specification limits.
13. Perform temperature cycling (-40°C to +85°C) and re-measure.
14. Document results and generate summary report.

5. ACCEPTANCE CRITERIA

- Insertion loss at center frequency shall be <= 3.7 dB
- Return loss in passband shall be >= 17.0 dB
- Out-of-band rejection at +/- 446 MHz offset >= 24.0 dB
- Group delay variation across passband shall be <= 6 ns
- Temperature drift over -40°C to +85°C shall be <= 18.6 MHz
- Power handling shall be >= +32 dBm at 1 dB compression
- ESD tolerance >= 2000 V (HBM)
- Die yield on qualification lot >= 95%

6. TEST RESULTS

Measurement Date: 2025-03-14
Devices Tested: 46
Insertion Loss (meas): 3.39 dB
Return Loss (meas): 19.1 dB
Rejection (meas): 27.3 dB
Isolation (meas): 49.0 dB
Q Factor (meas): 3130
Group Delay Variation: 3.0 ns
Power Handling: +26 dBm
Die Yield: 90.8%

Result Classification: NEEDS REVIEW

7. OBSERVATIONS AND FINDINGS

- a) The FBAR (Film Bulk Acoustic Resonator) design demonstrated below-target performance for Band 1 (2100 MHz).
- b) Insertion loss is marginally above target; electrode thickness optimization recommended.
- c) Rejection at near-band offset requires additional resonator tuning.
- d) Group delay flatness improved compared to previous revision.
- e) Cross-talk between adjacent filter channels was below -55 dB.

8. CORRECTIVE ACTIONS

- 1. Review near-band rejection margin for worst-case temperature.
- 2. Re-run EM simulation with refined mesh for better accuracy.
- 3. Schedule follow-up measurement after design tweak.

9. SIGN-OFF

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